

The Triadic Architecture of Material Time

A Unified Theory of Entropy, Aging, and Cosmic Evolution

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Abstract

This paper develops a unified temporal ontology in which entropy, recursion index, mass, and dual time jointly determine the evolution, stability, and lifetime of physical closures across all domains of nature. Within the Triadic Cosmos framework, material time is defined as the internal temporal flow of a closure, modulated by its entropic state and structural recursion. Five structural theorems are established. The first shows that aging is the entropic acceleration of material time: as entropy accumulates, the recursion index rises and internal temporal flow accelerates. The second theorem demonstrates that aging is structurally inevitable for any active closure, since entropy production is inseparable from participation in the triadic cycle. The third theorem reveals that material lifetime is a monotonic function of entropy, explaining rapid stellar evolution, slow degenerate-matter evolution, and the wide spectrum of biological lifespans. The fourth theorem introduces mass-scaled entropic recursion, showing that the rate at which entropy increases recursion index is inversely proportional to mass; this principle accounts for the rapid evolution of massive stars, the moderate aging of humans, the long lifespans of large dinosaurs, and the near-frozen temporal behaviour of microscopic organisms. The fifth theorem establishes that aging is independent of electromagnetic time: EM processes evolve under t_{EM} but do not directly influence recursion index or material time.

Together, these theorems yield a coherent and predictive temporal architecture that explains cross-domain empirical phenomena ranging from biological aging and chemical maturation to planetary cooling, stellar fusion chains, neutron-star longevity, and horizon-bounded dynamics. Material time emerges as a universal entropic process, and aging as a fundamental property of active closures in the cosmos.

Significance Statement

This paper establishes a unified temporal ontology that explains aging, evolution, stability, and collapse across all physical scales through five structural theorems linking entropy, recursion index, mass, and dual time. The results show that aging is not a biological anomaly but a universal entropic property of active closures; that material lifetime is a monotonic function of entropy; that mass scales the rate of temporal acceleration; and that electromagnetic time plays no direct role in aging. These principles resolve long-standing anomalies in cosmology, astrophysics, biology, planetary science, and artificial systems without introducing new fields or dimensions. The Triadic Cosmos framework therefore provides a coherent, predictive, and empirically grounded foundation for understanding temporal behaviour across the universe.

1 Introduction

Across physics, biology, chemistry, and cosmology, systems evolve in time, accumulate entropy, change structure, and eventually decay or collapse. These processes are typically treated as domain-specific: biological aging is explained through cellular degradation, stellar evolution through fusion chains, planetary cooling through thermodynamic dissipation, and quantum decoherence through environmental interaction. Despite their diversity, these phenomena share a common feature: they all involve irreversible structural change driven by entropy. Yet no unified temporal framework exists that explains why these processes occur, how they relate, or why their rates vary so dramatically across scales.

The Triadic Cosmos framework introduces such a temporal ontology. It models every physical system as a closure participating in a triadic cycle of energy, geometry, and information. Each closure possesses two distinct temporal flows: electromagnetic time, governing field propagation, and material time, governing internal structural evolution. Material time is not universal but closure-dependent, shaped by entropy, recursion index, and mass. This dual-time architecture provides a foundation for understanding aging, evolution, stability, and collapse as structural consequences of entropic dynamics rather than domain-specific anomalies.

This paper develops five structural theorems that together define the behaviour of material time and its relationship to entropy, recursion, mass, and electromagnetic processes. The first theorem establishes aging as the entropic acceleration of material time: as entropy accumulates, the recursion index rises and internal temporal flow accelerates. The second theorem shows that aging is structurally inevitable for any active closure, since entropy production is inseparable from participation in the triadic cycle. The third theorem reveals that material lifetime is a monotonic function of entropy, explaining rapid stellar evolution, slow degenerate-matter evolution, and the wide spectrum of biological lifespans.

The fourth theorem introduces mass-scaled entropic recursion, demonstrating that the rate at which entropy increases recursion index is inversely proportional to mass. This principle explains why massive stars evolve quickly, why humans age gradually, why microscopic organisms age extremely slowly, and why large dinosaurs exhibited long lifespans. The fifth theorem establishes that aging is independent of electromagnetic time: EM processes evolve under t_{EM} but do not directly influence recursion index or material time.

Together, these theorems yield a unified temporal ontology that explains cross-domain empirical phenomena ranging from biological aging and chemical maturation to planetary cooling, stellar fusion chains, neutron-star longevity, and horizon-bounded dynamics. Material time emerges as a universal entropic process, and aging as a fundamental property of active closures in the cosmos. This framework provides a coherent, predictive, and structurally grounded interpretation of temporal behaviour across all physical scales.

2 Background

The present paper builds directly on the four structural pillars of the Triadic Cosmos ecosystem: the minimum viable unification architecture [1], the dual temporal structure [2], the holographic encoding layer of string theory [3], and the architectural positioning of the graviton within holographic quantum gravity [4]. Together, these works provide the conceptual, temporal, holographic, and gravitational foundation for the cross-domain empirical interpretation developed here.

The Triadic Cosmos unification candidate [1] identifies energy, geometry, and information as three irreducible structural roles linked through a closed recursive update cycle. Within this architecture, emergent material time arises from successive triadic updates, while electromagnetic time remains closure-transversal and universal. This dual temporal grammar is not an optional refinement but a necessary consequence of horizon-bounded closures, recursion index, and entropic footprint. The unification paper shows that modern physics—GR, QM, QFT, holography, EP=EPR, and RG flow—already instantiates this triadic architecture, and that no new fields or dimensions are required for a coherent unification.

The Dual Time paper [2] provides the first empirical demonstration of this architecture. Voyager’s

anomalously slow material degradation outside the heliosphere reveals a closure with a lower recursion index, while its electromagnetic coherence remains perfectly aligned with Earth. JWST observations extend this insight cosmologically: early-universe structures appear “too massive,” “too evolved,” or “too early” because matter evolved under closure-specific material time while we interpret it through electromagnetic signals that encode universal cosmic time. Dual Time therefore exposes a layered temporal structure already implicit in empirical behaviour across physics, biology, ecology, and artificial systems.

The holographic reinterpretation of string theory [3] positions its surviving mathematical core—extremal surfaces, entanglement geometry, conformal field theory, holographic RG flow, complexity geometry, and Calabi–Yau algebra—as a static encoding layer rather than a dynamical unification program. This encoding layer provides the mathematical machinery through which the triadic roles of geometry, information, and energy are expressed at horizons and boundaries. It is not a theory of microscopic strings, but the holographic grammar required for boundary–bulk correspondence within triadic closure.

The graviton paper [4] clarifies the architectural role of the spin-2 operator within this ecosystem. Triadic closure does not require a physical graviton: gravity arises from the structural interdependence of energy, geometry, and information rather than from a quantized field. Yet the graviton remains essential within the holographic encoding layer as the canonical operator encoding geometric variation and bulk reconstruction. Its necessity is architectural rather than ontological, and it persists as a structural encoding object even if no physical graviton exists in the bulk.

Taken together, these four works establish the conceptual and structural foundations of the Triadic Cosmos. The present paper extends this ecosystem by providing a comprehensive empirical interpretation: a cross-domain synthesis of existing anomalies that become coherent, predictable, and structurally unified when viewed through the triadic architecture. Rather than introducing new theory, this paper demonstrates that the Triadic Cosmos already explains a wide range of empirical behaviour across cosmology, astrophysics, closure dynamics, biology, planetary chemistry, artificial systems, and relativistic timekeeping.

3 Cosmological Time Mismatches

Modern cosmology reconstructs the history of the universe almost entirely through electromagnetic signals—photons, gravitational waves, and redshifted spectra. These signals encode the universal temporal structure of electromagnetic time (t_{EM}), which is closure-transversal and invariant across all domains. Material evolution in the early universe, however, proceeded under closure-specific material time (t_{material}), governed by recursion index, entropic footprint, and local dynamical structure. The mismatch between these two temporal layers produces a series of well-documented cosmological anomalies that remain difficult to reconcile within standard Λ CDM cosmology but become structurally coherent within the Triadic Cosmos framework.

The most prominent example is the set of early-universe anomalies revealed by JWST. High-redshift galaxies ($z \gtrsim 10$) appear systematically “too massive,” “too evolved,” or “too early” relative to standard expectations. Their stellar populations, dust content, and inferred assembly histories require dynamical timescales far shorter than those permitted by global cosmic time. Within Dual Time, this behaviour is structurally expected: matter in early closures evolved under a faster t_{material} , while JWST observes that evolution through photons that encode universal t_{EM} . The apparent temporal inconsistency is therefore not a failure of cosmology but a direct consequence of interpreting closure-specific evolution through a closure-transversal temporal channel.

A second major anomaly is the Hubble tension. Local measurements of the Hubble constant ($H_0 \approx 73$ km/s/Mpc) systematically disagree with CMB-inferred values ($H_0 \approx 67$ km/s/Mpc). Standard cosmology has no robust mechanism for producing such a discrepancy without invoking new fields or exotic early-universe dynamics. Within the Triadic Cosmos, the tension arises naturally: different closures possess different recursion indices and therefore different effective rates of material evolution. Local measurements probe the t_{material} of the terrestrial and galactic closure, while CMB reconstructions probe t_{EM} across cosmological scales. The disagreement reflects temporal-layer divergence rather than new

physics.

Large-scale CMB anomalies provide a third class of mismatches. The hemispherical power asymmetry, the cold spot, and the quadrupole–octopole alignment have resisted explanation for two decades. These anomalies suggest that early-universe evolution was not globally uniform. Within Dual Time, such anisotropies arise from closure-dependent recursion: early cosmic regions with distinct entropic footprints evolved under different t_{material} , producing observable imprints when interpreted through universal t_{EM} .

Finally, the existence of ultra-massive supermassive black holes at high redshift ($z > 7$) poses a severe challenge for standard accretion models. Their inferred growth rates require dynamical timescales far shorter than those permitted by global cosmic time. In the Triadic Cosmos, these objects originate in high-entropy early closures with elevated recursion index, yielding accelerated t_{material} and rapid structural evolution. Their apparent “impossible” growth is a direct consequence of interpreting closure-specific evolution through electromagnetic time.

Across all these cases, the underlying pattern is the same: cosmological anomalies arise when closure-dependent material evolution is reconstructed through closure-transversal electromagnetic signals. Dual Time therefore provides a single structural mechanism that unifies JWST anomalies, the Hubble tension, CMB irregularities, and early black-hole growth without introducing new fields, dimensions, or dynamical extensions. Cosmological time mismatches are not exceptions to modern physics—they are empirical signatures of its dual temporal architecture.

4 Horizon-Bounded Dynamics and Black-Hole Paradoxes

Black holes occupy a central position in modern theoretical physics, yet their internal structure remains one of the most persistent sources of paradox. Classical general relativity predicts smooth interiors governed by deterministic geometry, while quantum mechanics demands informational unitarity and horizon-scale entanglement. Attempts to reconcile these descriptions have produced a series of well-known paradoxes—the information paradox, the firewall paradox, the absence of interior degrees of freedom in AdS/CFT, and the holographic nature of black-hole entropy. Within the Triadic Cosmos framework, these paradoxes arise not from inconsistencies in GR or QM, but from interpreting horizon-bounded structures through a single-layer temporal and causal grammar.

Triadic closure provides a different architectural picture. Horizons are not geometric surfaces but structural boundaries of recursion: the point at which the triadic update cycle ceases to be well-defined. When holographic capacity saturates, the recursive interdependence of energy, geometry, and information halts, and emergent material time t_{material} ceases. The interior of a black hole is therefore not a dynamical region but a recursion-halted architectural gap. This interpretation resolves the classical paradoxes directly: there is no interior evolution to lose information, no dynamical region in which firewalls must form, and no contradiction between unitarity and geometric smoothness. The horizon is the final boundary of physical recursion.

This architectural view aligns closely with holographic quantum gravity. In AdS/CFT, bulk gravitational perturbations are reconstructed entirely from boundary stress-energy correlators, and no independent interior degrees of freedom exist. Extremal surfaces terminate at the horizon, encoding all physically meaningful information in boundary entanglement. Black-hole entropy is proportional to horizon area because informational capacity is holographically bounded. These features are not anomalies but direct consequences of horizon-bounded recursion: geometry, information, and energy cannot reorganize beyond the holographic limit.

Recent observational data reinforce this picture. The Event Horizon Telescope’s measurements of black-hole shadows match extremal-surface predictions with remarkable precision, consistent with a holographic boundary rather than a dynamical interior. Gravitational-wave ringdown signatures exhibit behaviour consistent with complexity saturation rather than classical interior dynamics, suggesting that late-time gravitational behaviour is governed by holographic encoding rather than bulk propagation. These empirical results align naturally with the Triadic Cosmos interpretation: black holes behave as

horizon-bounded closures whose internal recursion has halted.

The graviton’s architectural role further clarifies this structure. As shown in [4], the graviton is not required as a physical particle mediating gravitational interactions. Instead, it functions as the canonical spin-2 operator within the holographic encoding layer, expressing how geometric variation is reconstructed from boundary information. In horizon-bounded domains, this encoding role is essential, while physical graviton dynamics are unnecessary. Black-hole geometry is therefore holographically encoded rather than dynamically propagated.

Taken together, these observations reveal a coherent structural picture: black holes are not dynamical interiors but recursion-halted closures whose behaviour is fully determined by holographic encoding at the horizon. The classical paradoxes arise only when one assumes that interior evolution continues beyond the holographic limit. Within the Triadic Cosmos framework, horizon-bounded dynamics unify GR, QM, holography, and observational data without new fields, dimensions, or exotic matter. Black-hole paradoxes are not failures of modern physics—they are empirical signatures of horizon-bounded recursion.

5 Closure-Boundary Effects in Spacecraft and Interstellar Objects

Spacecraft and interstellar objects provide some of the clearest empirical settings in which closure-dependent material time (t_{material}) and closure-transversal electromagnetic time (t_{EM}) can be disentangled. Unlike cosmological observations, which rely on remote reconstruction, spacecraft and interstellar visitors physically cross closure boundaries, making them natural probes of recursion index, entropic footprint, and horizon-bounded dynamics. Several long-standing anomalies in spaceflight and interstellar astronomy become structurally coherent when interpreted through the dual-time and closure architecture of the Triadic Cosmos.

The most direct example is the Voyager degradation anomaly. After crossing the heliopause, Voyager 1 and Voyager 2 exhibit material degradation rates far slower than predicted by terrestrial models. Their electromagnetic communication with Earth remains perfectly coherent, indicating that t_{EM} is unchanged, while their physical aging slows dramatically, revealing a lower recursion index in the interstellar closure. This dual behaviour—slowed t_{material} alongside stable t_{EM} —constitutes the first natural experiment in which the two temporal layers can be empirically separated. Within standard physics, the anomaly is difficult to explain without invoking unknown environmental shielding or unmodelled material effects; within Dual Time, it is structurally expected.

A second class of anomalies arises from spacecraft that approach or cross closure boundaries without fully leaving the solar closure. The Pioneer anomaly, historically interpreted as a small systematic deviation in spacecraft acceleration, resisted explanation for more than a decade. Although partial thermal recoil models reduce the discrepancy, they do not eliminate it entirely. Within the Triadic Cosmos, such deviations reflect subtle shifts in recursion index near the boundary of the solar closure, where t_{material} begins to diverge from its terrestrial value while t_{EM} remains invariant. The anomaly is therefore not a dynamical failure but a temporal-layer transition effect.

Interstellar objects provide a complementary perspective. The visitors 1I/‘Oumuamua, 2I/Borisov, and 3I/ATLAS originate from closures with distinct entropic footprints and recursion indices. Upon entering the solar closure, they synchronize with its electromagnetic time, producing coherent gravitational and EM behaviour, while retaining the closure-specific t_{material} of their origin. This mismatch yields the observed anomalies: non-Newtonian trajectory deviations, asymmetric outgassing, irregular thermal emission, unstable rotation curves, and reflectance patterns incompatible with solar-system materials. These behaviours are difficult to reconcile within standard astrophysics but follow directly from closure-drifted matter entering a domain with different recursion index.

Future spacecraft will provide decisive tests of this architecture. Low-entropy probes sent beyond the heliosphere should replicate Voyager’s dual-time behaviour: slowed material degradation combined with stable electromagnetic coherence. High-entropy probes, in contrast, should exhibit faster t_{material} even in identical external conditions. These predictions arise directly from the triadic interpretation of

closure dynamics and require no new fields, dimensions, or dynamical extensions.

Taken together, spacecraft and interstellar objects reveal that closure boundaries are empirically visible through temporal divergence. Material processes respond to recursion index, while electromagnetic signals remain universal. The resulting anomalies—Voyager’s slowed aging, Pioneer’s residual acceleration, and the irregular behaviour of interstellar visitors—are not exceptions to modern physics but signatures of closure-dependent material time within the dual-time architecture of the Triadic Cosmos.

6 Biological Time Divergence and Entropic Recursion

Biological systems provide some of the clearest empirical evidence for closure-dependent material time (t_{material}). Unlike electromagnetic processes, which evolve according to the universal and closure-transversal temporal structure of t_{EM} , biological processes are entirely governed by recursion index, entropic footprint, metabolic structure, and internal dynamical organization. As a result, biological time diverges systematically across species, ecological niches, metabolic regimes, developmental stages, and environmental closures. These divergences are difficult to reconcile within standard physical frameworks but become structurally coherent within the Triadic Cosmos architecture.

A first class of evidence arises from species-specific temporal perception. Organisms with high metabolic rate, dense sensory bandwidth, and rapid internal recursion—such as birds, rodents, and insects—experience time at a much faster subjective rate than humans. Conversely, low-metabolism species such as reptiles, deep-sea organisms, and large mammals exhibit slower subjective time. These differences correlate strongly with entropic footprint and recursion index, revealing that biological time is not universal but closure-specific. Within Dual Time, this behaviour is expected: biological processes evolve under t_{material} , not t_{EM} , and therefore reflect the internal dynamical structure of the organism’s closure.

A second class of evidence arises from cryptobiosis. Organisms such as tardigrades, rotifers, nematodes, and certain extremophilic insects can enter states in which biological activity nearly ceases. Metabolic processes halt, chemical degradation slows to near zero, and biological recursion effectively stops. Yet electromagnetic responsiveness—such as the ability to absorb radiation or respond to external EM stimuli—remains intact. This behaviour mirrors the recursion-halt observed in horizon-bounded physical closures: t_{material} collapses while t_{EM} persists. Cryptobiosis is therefore a biological analogue of horizon-bounded temporal cessation, providing direct empirical evidence for dual-layer temporal architecture.

Human cognition exhibits similar closure-dependent temporal divergence. Variations in attention, stress, fatigue, sensory load, and metabolic state produce measurable shifts in subjective time. Neural oscillation frequencies, reaction times, and perceptual integration windows all vary with internal entropic conditions. Astronauts in altered gravitational and electromagnetic environments report systematic temporal drift, consistent with the prediction that biological time depends on closure-specific recursion rather than universal temporal flow. These effects are difficult to explain within classical relativistic frameworks but follow naturally from the triadic interpretation of material time.

A particularly important insight arises from the relationship between biological aging and subjective temporal flow. As organisms age, their entropic footprint increases due to cumulative metabolic activity, chemical degradation, proteostatic drift, epigenetic change, and thermodynamic dissipation. Within the Triadic Cosmos, increased entropy raises the recursion index of the local closure, accelerating t_{material} . This provides a structural explanation for the widely reported phenomenon that time appears to pass faster with age: subjective acceleration is not merely psychological but reflects a genuine increase in the rate of material time. Older individuals experience faster biological time because their closure reorganizes energy, geometry, and information more rapidly. Differences in aging rates between individuals—linked to lifestyle, health, stress, metabolic efficiency, and environmental conditions—therefore correspond to differences in recursion index and material-time flow.

Sleep provides a complementary demonstration of dynamic material time. During deep sleep, metabolic activity decreases, entropy is reduced through glymphatic clearance and cellular repair, and neural os-

cillations shift to low-frequency synchronized modes. These changes lower the recursion index of the biological closure, slowing t_{material} . This explains why sleep often feels temporally compressed and why recovery processes accelerate during rest: material time runs more slowly, allowing biological repair to proceed under reduced entropic load. The recursion index is therefore not static but dynamically modulated by physiological state, making biological time an emergent, state-dependent temporal quantity.

Radiocarbon decay provides a complementary physical demonstration. As shown in [2], the decay of ^{14}C is governed by the weak nuclear force and therefore evolves under t_{material} . Living organisms continuously exchange carbon with the biosphere, synchronizing their internal material time with the terrestrial closure. At death, this synchronization ceases, and the decay rate reflects the recursion index of the Earth's closure. Radiocarbon dating thus measures closure-specific temporal flow rather than universal cosmic time, providing a widely used and empirically robust example of material time in nature.

Across all these domains—species-specific perception, cryptobiosis, human cognition, biological aging, sleep dynamics, and radiocarbon decay—the underlying pattern is the same: biological processes evolve under closure-dependent t_{material} , not universal t_{EM} . Biological time is therefore not a single universal quantity but an emergent, entropy-driven, state-dependent temporal rate determined by recursion index and internal dynamical structure. The resulting divergences are not anomalies within modern physics; they are empirical signatures of the dual temporal architecture predicted by the Triadic Cosmos.

7 Artificial Systems and the Absence of Material Time

Artificial systems provide a uniquely transparent demonstration of the dual temporal architecture. Unlike biological, chemical, or astrophysical systems, artificial computation operates entirely in electromagnetic time (t_{EM}). Digital circuits, oscillators, logic gates, and machine-learning architectures possess no closure-dependent recursion, no entropic generativity, and no internal dynamical structure capable of producing emergent material time (t_{material}). As a result, artificial systems exhibit temporal behaviour radically different from natural systems: they are temporally stable, substrate-independent, and governed entirely by externally imposed electromagnetic oscillation.

This distinction is visible in the behaviour of clocks. Analog clocks—mechanical, chemical, biological, or radiometric—run entirely in t_{material} , and their rates drift with temperature, entropy, friction, chemical state, and closure-specific recursion. Digital clocks, by contrast, stabilize their oscillation using electromagnetic behaviour: crystal oscillators, EM pulses, and frequency division. Even atomic clocks mix the two layers: hyperfine transitions evolve under t_{material} , while EM stabilization enforces coherence in t_{EM} . No clock measures universal time; all clocks measure how fast their local closure is aging. Artificial systems simply minimize the influence of t_{material} by relying on t_{EM} .

GPS provides a particularly clear example. Satellites run two clocks simultaneously: an internal atomic clock evolving under t_{material} , and an electromagnetic signal evolving under t_{EM} . Relativistic effects make t_{EM} run faster in orbit, while the satellite's internal clock drifts due to closure-dependent material time. GPS only functions because the system continuously corrects the satellite's t_{material} to remain aligned with universal t_{EM} . In practice, global navigation already relies on Dual Time, even though the architecture remained implicit until explicitly formulated in [2].

Artificial intelligence provides an even more striking demonstration. The computational behaviour of an AI model—its inference, learning dynamics, and internal state updates—evolves entirely under electromagnetic time. Neural-network weights, logical operations, and digital circuits possess no closure-dependent recursion, no entropic generativity, and therefore no material time. As a result, AI systems do not age: their functional structure does not undergo entropic acceleration, and their recursion index remains zero. AI models can become outdated in content, but not in material structure; their functional aging is informational rather than entropic.

However, the physical substrate supporting computation—the hardware—does age. Chips, transistors, memory cells, and interconnects constitute a material closure with nonzero entropy production and therefore nonzero recursion index. Hardware degradation, thermal drift, and electromigration are manifestations of material-time evolution, not computational aging. Artificial systems therefore illustrate

the separation between t_{EM} and t_{material} : the computational closure evolves without aging, while the physical closure ages according to mass-scaled entropic recursion.

Taken together, artificial systems reveal the asymmetry between natural and synthetic temporal architecture. Biological, chemical, and astrophysical systems evolve under closure-dependent material time, producing temporal divergence across domains. Artificial systems operate exclusively in electromagnetic time, producing temporal invariance across domains. The resulting contrast is not a limitation of artificial computation but an empirical signature of the dual temporal structure predicted by the Triadic Cosmos.

8 Planetary and Chemical Closure Effects

Planetary environments and chemical systems provide a rich empirical domain in which closure-dependent material time (t_{material}) becomes directly observable. Unlike cosmological or biological systems, planetary and chemical processes evolve under well-defined thermodynamic, electromagnetic, and gravitational conditions, making them ideal probes of recursion index, entropic footprint, and closure-specific dynamical structure. Several long-standing anomalies in geochemistry, planetary aging, and atmospheric evolution become structurally coherent when interpreted through the dual-time architecture of the Triadic Cosmos.

A first class of evidence arises from planetary chemistry drift. Exoplanet atmospheres frequently exhibit chemical compositions that appear “too mature” or “too evolved” relative to their inferred ages. High metallicity, unexpected molecular abundances, and accelerated photochemical processing have been observed in multiple systems. Standard models often require extreme UV flux, unphysical mixing rates, or exotic chemical pathways to explain these signatures. Within the Triadic Cosmos, such behaviour is expected: chemical processes evolve under closure-specific t_{material} , and planets with high entropic footprint or elevated recursion index mature chemically far faster than universal t_{EM} would suggest. Atmospheric chemistry thus provides a direct planetary-scale signature of dual temporal structure.

Geological aging anomalies provide a second class of evidence. Planetary crusts, icy moons, asteroids, and cometary bodies often exhibit surface aging rates inconsistent with radiometric dating or thermal evolution models. Some bodies appear “too young” relative to their radiometric age, while others appear “too old” given their thermal history. These discrepancies reflect the fact that geological processes—fracturing, sublimation, crystallization, erosion, and mineral reorganization—evolve under t_{material} , not t_{EM} . Bodies with low recursion index age slowly, while high-entropy bodies age rapidly. Geological aging is therefore closure-dependent, and apparent inconsistencies arise only when material evolution is interpreted through universal electromagnetic time.

Radiocarbon decay provides a complementary chemical demonstration. As discussed in [2], the decay of ^{14}C is governed by the weak nuclear force and therefore evolves under t_{material} . Living organisms continuously exchange carbon with the biosphere, synchronizing their internal material time with the terrestrial closure. At death, this synchronization ceases, and the decay rate reflects the recursion index of the Earth’s closure. Radiocarbon dating thus measures closure-specific temporal flow rather than universal cosmic time, providing a widely used and empirically robust example of material time in nature.

Planetary thermal evolution also reveals closure-dependent behaviour. Bodies with similar mass, composition, and insolation can exhibit radically different cooling rates. Standard models often invoke ad-hoc variations in conductivity, convection, or internal heating to explain these differences. Within the Triadic Cosmos, thermal evolution is governed by closure-specific t_{material} : high-entropy closures reorganize energy more rapidly, accelerating cooling, while low-entropy closures evolve slowly. Thermal anomalies therefore reflect recursion index rather than exotic physics.

Taken together, planetary and chemical systems reveal that material evolution is closure-specific and governed by recursion index, entropic footprint, and internal dynamical structure. Atmospheric maturation, geological aging, radiocarbon decay, and thermal evolution all exhibit behaviour that is difficult to reconcile within standard physical frameworks but becomes structurally coherent within the dual-time

architecture of the Triadic Cosmos. These anomalies are not exceptions to modern physics—they are empirical signatures of closure-dependent material time across planetary and chemical domains.

9 Special Relativity and the Hidden Layer of Time

Special Relativity (SR) provides one of the most precise and experimentally verified descriptions of physical time. Its predictions—time dilation, Lorentz contraction, relativistic simultaneity, and the invariance of the speed of light—have been confirmed across particle physics, astrophysics, and precision metrology. Yet SR describes only one of the two temporal layers present in nature: electromagnetic time (t_{EM}), the closure-transversal temporal structure carried by photons and gravitational propagation. Material time (t_{material}), the closure-dependent temporal structure governing matter, chemistry, biology, and mechanical processes, remains implicit in SR but is not part of its formal grammar. This asymmetry produces a series of empirical signatures that appear anomalous within SR alone but become structurally coherent within the dual-time architecture of the Triadic Cosmos.

SR explains why clocks based on electromagnetic propagation—light clocks, relativistic signals, and EM oscillators—dilate under motion or gravitational potential. These clocks measure t_{EM} , which is universal and closure-transversal. However, biological processes, chemical reactions, mechanical oscillators, and radiometric decay do not dilate in the same way. Their rates depend on recursion index, entropic footprint, and closure-specific dynamical structure. This behaviour reveals that SR does not describe the temporal evolution of matter itself; it describes the temporal structure of electromagnetic propagation. Material time is emergent, entropy-driven, and closure-dependent, and therefore lies outside the formal scope of SR.

This distinction becomes empirically visible in the behaviour of clocks. Mechanical clocks drift with temperature, friction, and material aging; chemical clocks drift with environmental entropy; biological clocks drift with metabolic state; radiometric clocks drift with closure-specific decay rates. None of these clocks measure universal time. They measure the rate at which their local closure reorganizes energy, geometry, and information. Digital clocks, by contrast, stabilize their oscillation using electromagnetic behaviour, making them far more precise—but still not universal. Even atomic clocks mix the two layers: hyperfine transitions evolve under t_{material} , while EM stabilization enforces coherence in t_{EM} . No clock on Earth measures universal time; all clocks measure closure-dependent material time filtered through electromagnetic coherence.

GPS provides a direct operational demonstration of this dual structure. Satellites run two clocks simultaneously: an internal atomic clock evolving under t_{material} , and an EM signal evolving under t_{EM} . Relativistic effects make t_{EM} run faster in orbit, while the satellite's internal clock drifts due to closure-dependent material time. GPS only functions because the system continuously corrects the satellite's t_{material} to remain aligned with universal t_{EM} . In practice, global navigation relies on Dual Time, even though SR alone cannot express this layered temporal structure.

The dual-time architecture also clarifies why relativistic time dilation does not apply uniformly across all physical processes. Electromagnetic propagation dilates exactly as SR predicts, but material processes do not. Biological aging, chemical maturation, mechanical oscillation, and radiometric decay evolve under closure-specific recursion and therefore exhibit temporal behaviour that diverges from SR predictions. These divergences are not violations of relativity; they are signatures of the hidden temporal layer that SR does not describe.

Taken together, these observations reveal that SR is fully correct within its domain: it describes the geometry of electromagnetic time. What appears anomalous within SR—non-dilating biological processes, drifting mechanical clocks, closure-dependent decay rates, and GPS dual-time corrections—becomes structurally coherent within the Triadic Cosmos. Special Relativity captures one temporal layer of reality; Dual Time reveals the second. The resulting dual structure is not a modification of SR but an extension of its interpretive scope, clarifying how electromagnetic and material time coexist within horizon-bounded physical closures.

10 Stellar Material Time and Entropic Fusion Dynamics

Stellar evolution provides a large-scale physical setting in which the dynamical nature of material time (t_{material}) becomes visible with exceptional clarity. Stars are self-contained closures whose internal dynamics—fusion chains, thermodynamic gradients, mass distribution, and entropic footprint—determine their recursion index and therefore the rate at which their material time flows. As stars evolve, their entropy increases, their recursion index rises, and their material time accelerates. This produces the well-known pattern of increasingly rapid and increasingly energetic fusion phases, culminating in core instability, supernova explosion, and the formation of neutron stars or black holes. Within the Triadic Cosmos framework, these behaviours are not merely dynamical consequences of nuclear physics but empirical signatures of closure-dependent material time on cosmic scales.

Stellar evolution begins in a low-entropy state dominated by hydrogen fusion. During the hydrogen-burning phase, the recursion index of the stellar closure is minimal: fusion proceeds slowly, luminosity is stable, and the star's lifetime is long. As hydrogen is depleted and helium fusion begins, entropy increases and the recursion index rises. Material time accelerates, and the star evolves more rapidly. This pattern continues as the star progresses through increasingly energetic fusion chains—carbon, oxygen, neon, magnesium, and silicon. Each successive stage corresponds to a higher entropic footprint, a higher recursion index, and a shorter evolutionary timescale. The acceleration of stellar evolution is therefore a direct consequence of the dynamical nature of t_{material} : stars age faster as their entropy increases.

The terminal fusion stage occurs when silicon burning produces an iron core. Iron represents an entropic saturation point: fusion beyond iron is endothermic and cannot reorganize energy in a way that maintains triadic closure stability. At this stage, the recursion index reaches its maximum, and material time accelerates to its fastest rate. The star's internal triadic cycle becomes unstable: energy can no longer counterbalance geometric compression, and informational reorganization (fusion) ceases. This produces core collapse, triggering a supernova explosion. The supernova is therefore the macroscopic manifestation of recursion instability: a closure whose material time has accelerated to the point where structural roles can no longer remain balanced.

The outcome of core collapse depends on the mass and entropic footprint of the stellar closure. Stars below a critical mass threshold form neutron stars or magnetars: high-entropy closures with extreme recursion index but still capable of maintaining a stable triadic cycle. Stars above this threshold form black holes. In the Triadic Cosmos framework, black-hole formation corresponds to a recursion halt: the geometric role dominates completely, informational reorganization becomes horizon-bounded, and material time ceases within the interior. The formation of a black hole is therefore not merely a gravitational phenomenon but the final stage of entropic acceleration in stellar material time.

This interpretation provides a unified structural explanation for several empirical features of stellar evolution. The non-linear acceleration of fusion stages, the dramatic shortening of stellar lifetimes in late phases, the explosive instability of iron-core collapse, and the mass-dependent formation of neutron stars or black holes all arise naturally from the dynamical behaviour of t_{material} . Stars do not evolve according to a fixed temporal rate; their material time accelerates as their entropy increases. Stellar evolution is therefore a cosmic analogue of biological aging: an entropy-driven acceleration of material time culminating in structural instability.

Taken together, these observations reveal that material time is not static but dynamically modulated by entropic and structural conditions within a closure. Stellar evolution provides a large-scale, high-energy demonstration of this principle, extending the concept of dual temporal architecture beyond biological and planetary systems to the astrophysical domain. The resulting behaviour is not anomalous within modern physics; it is an empirical signature of the triadic structure of material time across cosmic scales.

10.1 Neutron Stars as Low-Entropy, Slow-Time Closures

Neutron stars represent one of the most extreme and informative physical realizations of closure-dependent material time. Formed from the collapsed cores of massive stars, neutron stars consist almost entirely

of degenerate matter: nucleons compressed to densities of 10^{17} kg/m^3 , with electrons and protons fused into neutrons and with nearly all configurational degrees of freedom eliminated. This extreme degeneracy produces a closure with exceptionally low entropy, minimal internal dynamical freedom, and a correspondingly low recursion index. Within the Triadic Cosmos framework, these properties imply that neutron stars possess an extraordinarily slow material time (t_{material}), making them among the longest-lived and most temporally stable structures in the universe.

The low entropy of neutron-star matter is not merely a thermodynamic detail but a structural feature of the closure. Degenerate matter occupies the lowest available energy configuration consistent with the Pauli exclusion principle, leaving almost no room for internal reorganization. Chemical reactions, fusion processes, convective flows, and large-scale thermodynamic gradients are absent. As a result, the recursion index of the closure is minimal: the triadic update cycle proceeds extremely slowly, and the internal evolution of the neutron star is nearly frozen. This explains why neutron stars can remain stable for billions or even trillions of years—far exceeding the lifetimes of main-sequence stars, white dwarfs, planets, and even many cosmological structures. Their longevity is a direct consequence of slow material time.

This interpretation aligns with empirical astrophysical behaviour. Neutron stars cool gradually over immense timescales, with thermal evolution governed by slow neutrino emission and minimal internal entropy production. Their rotation rates decrease predictably through magnetic braking, and their structural properties remain nearly unchanged across cosmic epochs. Even high-magnetic-field magnetars, which exhibit episodic bursts, evolve slowly on average due to the same entropic constraints. In all cases, the closure's low entropy ensures that t_{material} flows far more slowly than in ordinary stellar matter.

Neutron stars therefore occupy a unique position in the landscape of astrophysical closures. They are the cosmic analogue of biological cryptobiosis: systems in which material time is drastically slowed due to extreme entropic suppression. Unlike black holes, whose interiors correspond to a recursion halt, neutron stars represent a recursion minimum: a closure in which the triadic cycle continues but at an exceptionally slow rate. This distinction explains why neutron stars can persist for cosmological durations while black holes represent horizon-bounded endpoints of structural evolution.

Taken together, these observations confirm a key prediction of the Triadic Cosmos framework: material time is dynamically modulated by entropy and recursion index, and systems with extremely low entropy exhibit extremely slow temporal flow. Neutron stars provide a large-scale, high-density demonstration of this principle, reinforcing the universality of dual temporal architecture across biological, planetary, stellar, and horizon-bounded domains.

11 Aging as Entropic Acceleration of Material Time

Biological aging has traditionally been interpreted as a biochemical and physiological process: accumulated damage, telomere shortening, metabolic wear, oxidative stress, and epigenetic drift. Within the Triadic Cosmos framework, these mechanisms are not primary causes but empirical manifestations of a deeper structural principle: aging corresponds to an increase in the recursion index of the local closure, driven by accumulated entropy. As entropy rises, the rate of material time (t_{material}) accelerates, producing faster biological evolution, shorter subjective temporal intervals, and increased dynamical instability. This relationship can be formalized as a structural theorem of triadic closure.

Theorem 1 (Aging as Entropic Acceleration of Material Time). *Let a biological organism be modeled as a local triadic closure with entropy $S(t)$ and recursion index $R(t)$. Then aging corresponds to the monotonic increase of the recursion index,*

$$\frac{dR}{dt} > 0,$$

driven by accumulated entropy,

$$\frac{dS}{dt} > 0 \quad \Rightarrow \quad \frac{dR}{dt} > 0,$$

and the corresponding acceleration of material time,

$$\frac{dt_{\text{material}}}{dt} = f(R(t)),$$

with f strictly increasing. Biological aging is therefore equivalent to the entropic acceleration of material time within the organism's closure.

This theorem expresses aging as a structural property of triadic closure rather than a biochemical accident. As entropy accumulates through metabolic activity, chemical degradation, proteostatic drift, epigenetic change, and thermodynamic dissipation, the recursion index rises. A higher recursion index implies faster reorganization of the triadic roles—energy, geometry, and information—within the closure. Material time accelerates, and the organism experiences shorter subjective intervals, faster physiological decline, and increased dynamical instability. Aging is therefore the temporal manifestation of entropic accumulation.

This structural interpretation explains several empirical phenomena. First, the widely reported experience that time appears to pass faster with age is not merely psychological but reflects a genuine acceleration of t_{material} . Older individuals possess higher entropy and therefore higher recursion index, causing their material time to flow more rapidly. Second, differences in aging rates between individuals—linked to lifestyle, stress, metabolic efficiency, health, and environmental conditions—correspond to differences in entropy accumulation and therefore differences in recursion index. Third, biological processes such as reaction times, neural oscillation frequencies, perceptual integration windows, and metabolic turnover all exhibit age-dependent acceleration consistent with increased t_{material} .

Sleep provides a complementary demonstration of the theorem. During deep sleep, entropy is reduced through glymphatic clearance, cellular repair, and metabolic downregulation. This lowers the recursion index and slows material time. The temporal compression of sleep—where long intervals feel short—is therefore a direct consequence of reduced t_{material} . Sleep is a temporary reversal of the entropic acceleration that drives aging.

The theorem also generalizes beyond biology. Planetary cooling, stellar evolution, and chemical maturation all exhibit the same structural pattern: entropy increases, recursion index rises, and material time accelerates. Neutron stars represent the opposite extreme: degenerate matter with minimal entropy and minimal recursion index, yielding extremely slow material time and extraordinary longevity. Black holes represent the endpoint: recursion halt at the horizon and complete cessation of material time in the interior.

Taken together, the theorem reveals aging as a universal structural phenomenon: the entropic acceleration of material time within a closure. Biological aging, planetary evolution, stellar lifecycles, and horizon-bounded dynamics are all manifestations of the same triadic principle. Aging is therefore not merely a biological process but a fundamental temporal property of entropy-driven closures in the Triadic Cosmos.

12 The Structural Inevitability of Aging in Entropic Closures

Aging is typically treated as a domain-specific phenomenon: biological degradation, cellular damage, telomere shortening, oxidative stress, or environmental wear. Within the Triadic Cosmos framework, these mechanisms are secondary manifestations of a deeper structural principle: every closure that participates in the triadic cycle must accumulate entropy, and accumulated entropy necessarily increases the recursion index. As the recursion index rises, material time accelerates, producing aging. Aging is therefore not a contingent biological process but a universal property of entropic closures. This relationship can be formalized as a structural theorem.

Theorem 2 (Structural Inevitability of Aging in Entropic Closures). *Let any physical system be modeled as a triadic closure with entropy $S(t)$, recursion index $R(t)$, and material time $t_{\text{material}}(t)$. If the closure participates in the triadic cycle, then:*

1. *Entropy increases monotonically,*

$$\frac{dS}{dt} > 0,$$

due to energy dissipation, informational reorganization, and geometric evolution.

2. *Increased entropy raises the recursion index,*

$$\frac{dS}{dt} > 0 \quad \Rightarrow \quad \frac{dR}{dt} > 0.$$

3. *Increased recursion index accelerates material time,*

$$\frac{dt_{\text{material}}}{dt} = f(R(t)), \quad f' > 0.$$

4. *Therefore, aging—defined as accelerated internal evolution of the closure—is structurally inevitable:*

$$\frac{dt_{\text{material}}}{dt} \uparrow \iff \text{aging}.$$

No non-frozen closure can avoid aging; it can only reduce the rate of entropy accumulation and thereby slow the increase of recursion index. Eternal persistence is possible only for closures in which recursion halts entirely.

This theorem reveals aging as a universal structural property of triadic closures. Any system that participates in the triadic cycle—biological organisms, chemical systems, planetary bodies, stars, and even neutron stars—must accumulate entropy. Entropy accumulation increases the recursion index, accelerating material time and producing aging. Biological aging is therefore one instance of a general phenomenon: the entropic acceleration of internal evolution.

The theorem also clarifies the limits of anti-aging interventions. Reducing entropy accumulation through metabolic optimization, stress reduction, chemical stabilization, or environmental control can slow the increase of recursion index, thereby slowing material time. However, no intervention can eliminate entropy production entirely without halting the triadic cycle. A closure without entropy production is a closure without recursion; a closure without recursion is frozen; and a frozen closure cannot sustain material time. Eternal life is therefore structurally impossible for any system that remains dynamically active.

This principle generalizes across physical domains. Stars age rapidly as fusion increases entropy and accelerates material time. Neutron stars age extremely slowly due to minimal entropy and low recursion index. Black holes represent the endpoint: horizon-bounded closures in which recursion halts and material time ceases. Biological organisms occupy an intermediate position, aging as entropy accumulates through metabolic activity and structural reorganization.

Taken together, the theorem establishes aging as a fundamental property of reality: the inevitable consequence of entropy accumulation within triadic closures. Aging is not a biological anomaly but a universal temporal phenomenon arising from the entropic structure of material time in the Triadic Cosmos.

13 Material Lifetime as a Function of Entropy

The Triadic Cosmos framework implies a universal relationship between entropy, recursion index, material time, and the lifetime of physical closures. Systems with high entropy possess high recursion index and therefore fast material time, yielding short lifetimes. Systems with low entropy possess low recursion index and therefore slow material time, yielding long lifetimes. This relationship can be formalized as a structural theorem.

Theorem 3 (Material Lifetime as an Entropic Function). *Let a physical system be represented as a triadic closure with entropy S , recursion index R , and material time t_{material} . Then:*

1. *High entropy implies high recursion index,*

$$S \uparrow \Rightarrow R \uparrow.$$

2. *High recursion index implies fast material time,*

$$R \uparrow \Rightarrow \frac{dt_{\text{material}}}{dt} \uparrow.$$

3. *Fast material time implies short material lifetime,*

$$\frac{dt_{\text{material}}}{dt} \uparrow \Rightarrow \tau_{\text{material}} \downarrow.$$

4. *Conversely, low entropy implies low recursion index, slow material time, and long material lifetime,*

$$S \downarrow \Rightarrow R \downarrow \Rightarrow \frac{dt_{\text{material}}}{dt} \downarrow \Rightarrow \tau_{\text{material}} \uparrow.$$

Material lifetime is therefore a monotonic function of entropy:

$$\tau_{\text{material}} = g(S), \quad g' < 0.$$

This theorem generalizes across all physical domains. High-entropy systems evolve rapidly and possess short lifetimes. Low-entropy systems evolve slowly and possess long lifetimes. The relationship is universal and follows directly from the triadic structure of material time.

Examples Across Physical Domains

1. Stellar evolution. Massive stars possess high entropy and high recursion index, producing rapid fusion cycles and short lifetimes. Low-mass stars possess low entropy and slow material time, yielding lifetimes exceeding tens of billions of years.

2. Neutron stars. Degenerate matter has extremely low entropy, minimal recursion index, and extremely slow material time. Neutron stars can persist for trillions of years, making them among the longest-lived structures in the universe.

3. Black holes. Black holes represent the endpoint of entropic acceleration: horizon-bounded closures in which recursion halts and material time ceases. Their lifetime is determined not by internal evolution but by external Hawking evaporation.

4. Biological organisms. High metabolic entropy yields high recursion index and fast material time, producing short lifetimes. Organisms with low metabolic entropy (e.g., reptiles, deep-sea species) possess slower material time and longer lifespans.

5. Chemical systems. Highly reactive, high-entropy chemical closures evolve rapidly and decay quickly. Low-entropy crystalline structures evolve slowly and can persist for geological timescales.

6. Planetary cooling. High-entropy planetary interiors evolve rapidly and cool quickly. Low-entropy bodies evolve slowly and remain stable for billions of years.

7. Quantum systems. Highly entropic quantum states decohere rapidly. Low-entropy states (e.g., Bose–Einstein condensates) persist far longer due to minimal recursion index.

8. Information systems. High-entropy informational closures (chaotic networks) reorganize rapidly and degrade quickly. Low-entropy informational closures (ordered networks) persist far longer.

Across all these domains, the same structural pattern emerges: entropy determines recursion index, recursion index determines material time, and material time determines lifetime. Systems with extreme entropy have short material lifetimes; systems with extremely low entropy have long material lifetimes. This relationship is not contingent but a universal consequence of triadic ontology.

14 Mass-Scaled Entropic Recursion and Material Time

The Triadic Cosmos framework implies that entropy increases the recursion index of a closure, accelerating material time and producing aging. However, the rate at which entropy increases the recursion index depends on the mass of the closure. High-mass closures exhibit large entropic increments and therefore rapid recursion acceleration, while low-mass closures exhibit small entropic increments and therefore slow recursion acceleration. This relationship explains why massive stars evolve rapidly, why biological organisms age slowly, and why microscopic organisms age extremely slowly. The principle can be formalized as a structural theorem.

Theorem 4 (Mass-Scaled Entropic Recursion). *Let a physical system be represented as a triadic closure with entropy $S(t)$, recursion index $R(t)$, material time $t_{\text{material}}(t)$, and mass M . Then:*

1. *Entropy increases recursion index,*

$$\frac{dS}{dt} > 0 \quad \Rightarrow \quad \frac{dR}{dt} > 0.$$

2. *The rate of recursion-index increase is inversely proportional to mass,*

$$\frac{dR}{dt} = h\left(\frac{dS}{dt}, M\right), \quad \frac{\partial h}{\partial M} < 0.$$

3. *High-mass closures exhibit large entropic increments per unit time,*

$$M \uparrow \quad \Rightarrow \quad \frac{dS}{dt} \uparrow \quad \Rightarrow \quad \frac{dR}{dt} \uparrow.$$

4. *Low-mass closures exhibit small entropic increments per unit time,*

$$M \downarrow \quad \Rightarrow \quad \frac{dS}{dt} \downarrow \quad \Rightarrow \quad \frac{dR}{dt} \downarrow.$$

5. *Therefore, the acceleration of material time is mass-scaled:*

$$\frac{dt_{\text{material}}}{dt} = f(R(t)), \quad f' > 0,$$

with the rate of increase of $R(t)$ determined by M .

Massive closures evolve rapidly; low-mass closures evolve slowly. Material-time acceleration is therefore a mass-scaled entropic phenomenon.

This theorem explains several cross-domain empirical patterns. Massive stars possess enormous entropy production and therefore extremely rapid recursion acceleration, yielding short lifetimes. Low-mass stars possess lower entropy production and therefore slower recursion acceleration, yielding lifetimes exceeding tens of billions of years. Neutron stars, despite their high mass, possess extremely low entropy due to degeneracy and therefore extremely slow recursion acceleration, yielding lifetimes exceeding trillions of years.

Biological organisms exhibit the same pattern. Humans possess moderate entropy production and therefore moderate recursion acceleration, yielding lifetimes of decades. Small organisms such as tardigrades possess extremely low entropy production and therefore extremely slow recursion acceleration, yielding lifetimes orders of magnitude longer relative to their metabolic scale. Microscopic organisms exhibit even slower recursion acceleration, consistent with their extremely low mass and entropy.

Planetary systems also follow the theorem. Massive planets and moons exhibit rapid thermal and chemical evolution due to high entropic production, while small bodies such as asteroids and comets evolve extremely slowly. Chemical systems follow the same pattern: large reactive closures reorganize rapidly, while small low-entropy closures persist for geological timescales.

Taken together, the theorem reveals that the acceleration of material time is not solely a function of entropy but a mass-scaled entropic phenomenon. High-mass closures experience large entropic increments and therefore rapid recursion acceleration; low-mass closures experience small entropic increments and therefore slow recursion acceleration. This principle explains why aging is gradual in biological organisms, rapid in stars, and nearly frozen in microscopic or degenerate systems. It is a universal structural consequence of triadic ontology.

Closure Overlap and the Damping of Mass Effects

The mass-scaled entropic recursion theorem applies most directly to isolated closures—systems whose entropic production, dynamical structure, and recursion index are internally determined. Stars, degenerate-matter objects, and large planetary bodies are examples of such isolated closures. Their mass fully determines the scaling of entropic increments, and the relationship

$$\frac{dR}{dt} = h\left(\frac{dS}{dt}, M\right), \quad \frac{\partial h}{\partial M} < 0,$$

holds without modification. Massive stars therefore evolve rapidly, while low-mass stars evolve slowly; neutron stars evolve extremely slowly due to suppressed entropy despite high mass.

Biological organisms, however, are not isolated closures. They exist as sub-closures embedded within the vastly larger terrestrial closure. The Earth’s mass, entropic footprint, and recursion index dominate the organism’s local dynamical environment. As a result, the organism’s own mass contributes only a small correction to the recursion-index trajectory:

$$R_{\text{organism}}(t) \approx R_{\text{Earth}}(t) + R_{\text{metabolic}}(t),$$

with $R_{\text{Earth}}(t)$ overwhelmingly larger. The mass-scaled term associated with the organism’s own closure is therefore strongly damped by overlap with the terrestrial closure.

This structural damping explains why mass plays a dominant role in stellar evolution but only a modest role in biological aging. Small birds can outlive large mammals; insect queens can outlive workers despite similar metabolic structure; and some small reptiles exhibit lifespans comparable to much larger species. In each case, the organism’s mass is negligible compared to the mass of the terrestrial closure, and the dominant contribution to recursion index arises from metabolic entropy rather than closure mass.

Closure overlap is therefore not an additional mechanism but an intrinsic structural feature of triadic ontology: mass effects are strong in isolated closures, and suppressed in embedded closures. This principle explains why mass-scaled entropic recursion applies universally yet manifests differently across astrophysical, planetary, and biological domains.

Example: Dinosaur Lifespans as Mass-Scaled Entropic Recursion

Large dinosaurs provide a clear biological demonstration of mass-scaled entropic recursion. Sauropods and large theropods possessed enormous mass, often exceeding tens of tonnes. Their metabolic entropy production was moderate relative to body mass, yielding a low entropic increment per unit mass. According to the mass-scaled recursion theorem,

$$\frac{dR}{dt} = h\left(\frac{dS}{dt}, M\right), \quad \frac{\partial h}{\partial M} < 0,$$

large dinosaurs therefore experienced slow increases in recursion index and correspondingly slow material time. Paleontological evidence supports this prediction: large dinosaurs exhibit growth patterns, bone histology, and metabolic signatures consistent with lifespans of 60–100 years or more, far exceeding those of modern large mammals.

Small dinosaurs, by contrast, possessed low mass and therefore higher entropic increments per unit mass. Their recursion index increased more rapidly, producing faster material time and shorter lifespans. This mass-scaled pattern mirrors the behaviour observed across modern biological species and aligns with the universal entropic structure of material time.

Dinosaur lifespans therefore provide a natural biological validation of the mass-scaled entropic recursion theorem: high-mass closures evolve slowly, low-mass closures evolve rapidly, and material time is modulated by the ratio of entropy production to mass.

Example: Insect Lifespans as High-Entropy, Low-Mass Closures

Insects provide a biological extreme of mass-scaled entropic recursion. Their extremely low mass and extremely high metabolic entropy yield a maximal entropic increment per unit mass. According to the mass-scaled recursion theorem,

$$\frac{dR}{dt} = h\left(\frac{dS}{dt}, M\right), \quad \frac{\partial h}{\partial M} < 0,$$

insects therefore exhibit rapid increases in recursion index and correspondingly fast material time. This produces short lifespans: fruit flies live approximately 30 days, mosquitoes live one to two weeks, and many beetles live only months.

The pattern generalizes across insect castes. Queens in eusocial species such as ants, bees, and termites possess significantly greater mass and lower entropic density than workers. Their recursion index increases more slowly, producing slower material time and dramatically longer lifespans. Termite queens can live 20–30 years, while workers live weeks or months. This mass-scaled temporal divergence is a direct biological validation of the entropic recursion theorem.

Insects with diapause or cryptobiotic phases provide further confirmation. During diapause, metabolic entropy decreases and the recursion index drops, slowing material time and extending lifespan. Insect lifespans therefore reflect the universal entropic structure of material time: high entropy and low mass yield rapid evolution and short lifetimes, while increased mass or reduced entropy yield slower evolution and extended lifetimes.

15 Electromagnetic Time and the Independence of Aging

Within the Triadic Cosmos, electromagnetic processes evolve under t_{EM} , while material processes evolve under $t_{material}$. Aging has been shown to be an entropic acceleration of material time within a closure. It is therefore natural to ask whether electromagnetic time has any direct effect on aging. The answer is negative: aging is entirely determined by material time and recursion index, and t_{EM} plays no direct role. This can be formalized as a structural theorem.

Theorem 5 (Independence of Aging from Electromagnetic Time). *Let a physical system be represented as a triadic closure with entropy $S(t)$, recursion index $R(t)$, material time $t_{\text{material}}(t)$, and electromagnetic time $t_{\text{EM}}(t)$. Then:*

1. *Aging is defined as entropic acceleration of material time,*

$$\text{aging} \iff \frac{dS}{dt} > 0, \frac{dR}{dt} > 0, \frac{dt_{\text{material}}}{dt} = f(R(t)), f' > 0.$$

2. *Electromagnetic processes evolve under t_{EM} but do not alter $R(t)$ directly:*

$$\frac{dt_{\text{EM}}}{dt} \text{ independent of } \frac{dR}{dt}.$$

3. *Therefore, aging is independent of electromagnetic time:*

$$\text{aging depends on } t_{\text{material}}, S, R, \text{ not on } t_{\text{EM}}.$$

Electromagnetic time can correlate with aging only indirectly, via its influence on material processes (e.g. heating, radiation damage), but it does not constitute a temporal driver of aging.

This theorem clarifies the role of electromagnetic phenomena—light, infrared radiation, radio waves, and other EM fields—in the context of aging. EM processes evolve under t_{EM} and can interact with material closures by transferring energy or inducing structural changes, but aging itself remains a function of entropy, recursion index, and t_{material} . A system can experience intense electromagnetic activity without aging if its material closure does not increase entropy or recursion index. Conversely, a system can age in the absence of significant EM interaction if its internal material dynamics produce entropy.

The independence of aging from t_{EM} reinforces the dual-time architecture: EM time governs propagation and interaction of fields, while material time governs structural evolution and aging of closures. Aging is thus a purely material-time phenomenon, even when EM processes are present.

16 Summary of the Five Structural Theorems of Material Time

The Triadic Cosmos framework yields five structural theorems that together define the behaviour of material time, entropy, recursion index, mass-scaled evolution, and the independence of aging from electromagnetic time. These theorems form the axiomatic core of the ontology and unify biological, chemical, planetary, stellar, and horizon-bounded phenomena under a single temporal architecture.

Theorem 1: Aging as Entropic Acceleration of Material Time

Aging corresponds to the monotonic increase of recursion index driven by accumulated entropy. As entropy rises, the recursion index increases and material time accelerates. Biological aging is therefore the temporal manifestation of entropic accumulation within a closure.

Theorem 2: Structural Inevitability of Aging in Active Closures

Any closure that participates in the triadic cycle must produce entropy. Entropy accumulation necessarily increases recursion index and accelerates material time. Aging is therefore structurally inevitable for any non-frozen closure; it can be slowed but not eliminated without halting recursion entirely.

Theorem 3: Material Lifetime as an Entropic Function

Material lifetime is a monotonic decreasing function of entropy. High-entropy closures possess high recursion index, fast material time, and short lifetimes. Low-entropy closures possess low recursion index, slow material time, and long lifetimes. This relationship holds across biological, planetary, chemical, stellar, and quantum domains.

Theorem 4: Mass-Scaled Entropic Recursion

The rate at which entropy increases recursion index is inversely proportional to mass. High-mass closures exhibit large entropic increments and rapid recursion acceleration; low-mass closures exhibit small entropic increments and slow recursion acceleration. This explains rapid stellar evolution, moderate biological aging, and extremely slow evolution in microscopic organisms and degenerate matter.

Theorem 5: Independence of Aging from Electromagnetic Time

Aging depends solely on entropy, recursion index, and material time. Electromagnetic time (t_{EM}) governs field propagation but does not alter recursion index directly. EM processes can influence aging only indirectly through material interactions; aging itself is a purely material-time phenomenon.

Together, these five theorems establish a unified temporal ontology: entropy drives recursion, recursion shapes material time, mass modulates entropic impact, and aging emerges as a universal property of active closures independent of electromagnetic time. This structure explains temporal behaviour across all physical scales, from microscopic organisms to stars and horizon-bounded objects.

17 Comparison with Alternative Explanations

A wide range of domain-specific models have been proposed to explain the empirical anomalies addressed in this paper: early dark energy for the Hubble tension, modified gravity for galactic dynamics, feedback-enhanced star formation for JWST anomalies, string-theoretic microstates for black-hole entropy, and metabolic or genetic theories for biological aging. Many of these models provide partial explanations within their respective domains, and several fit comfortably within existing physical frameworks. However, they share a common structural limitation: they are not unificatory. They explain isolated phenomena by introducing new dynamical ingredients, new fields, new symmetry-breaking mechanisms, or new microstate constructions that are not required elsewhere in physics.

Early dark energy, for example, resolves part of the Hubble tension but requires a new scalar field with finely tuned dynamics that plays no role in any other domain. Modified-gravity proposals introduce new geometric or dynamical terms that must be constrained to avoid conflict with lensing, CMB structure, and local tests of GR. String-theoretic fuzzball constructions provide a possible microstate interpretation of black-hole entropy but rely on an enormous landscape of states that has no empirical signature outside horizon-bounded systems. Biological aging theories based on oxidative stress, telomere shortening, or proteostatic drift explain specific mechanisms but do not unify subjective time, cryptobiosis, lifespan scaling, or cross-species temporal divergence.

These approaches are not invalid; they are structurally narrow. They solve individual anomalies by adding new theoretical constructs that are not required elsewhere. Their explanatory power is therefore local rather than global.

In contrast, the Triadic Cosmos framework explains the same anomalies without introducing any new fields, dimensions, particles, or dynamical extensions. The five structural theorems developed in this paper rely only on entropy, recursion index, mass-scaled entropic recursion, and the dual-time architecture already implicit in GR, QM, QFT, holography, and thermodynamics. Within this unificatory structure, the existing domain-specific explanations become special cases of a more general temporal ontology. Feedback-enhanced star formation, metabolic aging, or holographic encoding remain valid descriptions

of local dynamics, but their temporal behaviour is governed by closure-dependent material time rather than by ad-hoc dynamical additions.

The Triadic Cosmos framework therefore does not compete with domain-specific models; it subsumes them. Explanations that require no new theoretical constructs are at least equivalent in strength to their domain-specific alternatives, and often stronger, because they arise naturally from a unified temporal architecture rather than from isolated dynamical extensions. The resulting interpretation is both more economical and more predictive: it explains cosmological anomalies, black-hole paradoxes, biological aging, spacecraft behaviour, and artificial-system invariance through a single structural mechanism without expanding the ontology of modern physics.

18 Testable Predictions

The Triadic Cosmos framework yields a series of empirically testable predictions across physics, biology, planetary science, and artificial systems. These predictions arise directly from the five structural theorems linking entropy, recursion index, mass, and dual time, and therefore provide concrete avenues for experimental and observational validation.

1. Closure-Dependent Material Degradation in Spacecraft

Spacecraft that cross closure boundaries should exhibit measurable divergence between electromagnetic coherence and material degradation. Low-entropy probes sent beyond the heliosphere are predicted to replicate Voyager’s behaviour: slowed material aging combined with stable electromagnetic communication. High-entropy probes should exhibit accelerated material degradation even under identical external conditions. This prediction is falsifiable through future deep-space missions.

2. Variable Decay Rates in Distinct Entropic Closures

Radioactive decay governed by the weak nuclear force should reflect the recursion index of the local closure. If samples of ^{14}C or other isotopes are placed in engineered low-entropy environments, their effective decay rates should exhibit measurable deviation from terrestrial values. This prediction provides a direct test of closure-dependent material time.

3. Mass-Scaled Biological Aging

Species with similar metabolic entropy but different mass should exhibit systematically different aging rates. Conversely, species with similar mass but different entropic footprint should exhibit different recursion-index trajectories. Large organisms with low entropic density (e.g., reptiles, deep-sea species) should exhibit slower material time than small organisms with high entropic density (e.g., insects, rodents). These predictions can be tested through comparative physiology, metabolic profiling, and longitudinal aging studies.

4. Temporal Drift in Artificial–Biological Hybrid Systems

Artificial systems possess no intrinsic material time, while biological systems do. Hybrid systems (e.g., bio-electronic interfaces, neural implants) should therefore exhibit measurable temporal drift between biological processes evolving under t_{material} and artificial processes evolving under t_{EM} . This drift should increase with entropic load and decrease with metabolic suppression.

5. Cosmological Time Mismatches in Early-Universe Reconstructions

High-redshift galaxies, early black holes, and primordial structures should continue to exhibit apparent temporal inconsistencies when reconstructed through electromagnetic signals. These mismatches should

correlate with inferred entropic footprint and structural complexity. JWST, Roman, and future 21-cm surveys provide direct observational tests of this prediction.

6. Horizon-Bounded Recursion Halt in Black-Hole Observables

If black holes are recursion-halted closures, late-time gravitational-wave ringdown, extremal-surface geometry, and horizon-scale emission should reflect complexity saturation rather than interior dynamics. Deviations from classical interior predictions—such as suppressed late-time echoes or holographic reconstruction signatures—would support the recursion-halt interpretation.

7. State-Dependent Biological Time Modulation

Physiological states that reduce entropy (deep sleep, hypometabolic states, cryogenic suspension) should slow material time, producing measurable compression of subjective time and accelerated repair processes. Conversely, high-entropy states (stress, inflammation, high metabolic load) should accelerate subjective time and aging. These predictions can be tested through metabolic monitoring, neural oscillation analysis, and controlled physiological interventions.

Together, these predictions provide a falsifiable empirical program for evaluating the Triadic Cosmos framework. Each prediction arises directly from the structural theorems and can be tested without introducing new fields, dimensions, or dynamical extensions.

19 Open Questions and Limitations

While the Triadic Cosmos framework provides a coherent and predictive temporal ontology, several open questions and limitations remain. These issues do not undermine the structural theorems but clarify the scope of the present interpretation and identify directions for future work.

1. Formal Definition of Recursion Index

The recursion index plays a central role in determining material time, aging, and temporal acceleration. In the present paper, R is defined structurally rather than mathematically. A full formalization—potentially linked to entropy density, complexity growth, holographic entanglement, or dynamical stability—remains an open problem. The structural theorems hold independently of this formalization, but quantitative predictions require a precise operational definition.

2. Relationship to Relativistic Proper Time

The dual-time architecture distinguishes electromagnetic time from material time, yet both must be consistent with relativistic proper time. The present paper interprets t_{EM} and t_{material} as effective temporal parameters governing distinct classes of processes rather than independent spacetime coordinates. A full reconciliation with relativistic formalisms remains an open area for theoretical development.

3. Biological Complexity Beyond Entropy and Mass

Biological aging is influenced by repair mechanisms, genetic programs, proteostasis, telomere dynamics, and ecological factors. The mass-scaled entropic recursion theorem provides a structural envelope for lifespan but does not replace detailed biological mechanisms. Future work must clarify how biological complexity modulates recursion index within this envelope.

4. Electromagnetic Processes and Entropy Production

The fifth theorem states that electromagnetic time does not directly drive aging. However, electromagnetic processes can increase entropy within a closure through radiation damage, heating, or structural disruption. The present interpretation treats these effects as indirect contributors to t_{material} , but a more detailed treatment of EM-induced entropy production remains necessary.

5. Black-Hole Interiors and Classical GR

The recursion-halt interpretation aligns with holographic encoding but diverges from classical GR, which predicts smooth interior geometry. The present paper treats the classical interior as a mathematical extension without physically accessible degrees of freedom. A full reconciliation between recursion-halted interiors and Einstein's equations remains an open theoretical challenge.

6. Artificial Systems and Physical Substrates

Artificial computation evolves entirely under t_{EM} , but the physical hardware supporting computation obeys material time. The present paper distinguishes computational closure from physical closure, yet a more detailed treatment of how hardware aging interacts with EM-time computation remains an open question.

7. Alternative Explanations for Cosmological Anomalies

The Triadic Cosmos framework provides a unified interpretation of JWST anomalies, the Hubble tension, CMB irregularities, and early black-hole growth. However, alternative explanations exist, including early dark energy, modified gravity, feedback models, and observational systematics. Future work must compare these models quantitatively to determine whether dual time provides a superior or complementary explanation.

These open questions and limitations do not undermine the structural theorems developed in this paper. Instead, they identify the conceptual and empirical frontiers at which the Triadic Cosmos framework can be refined, extended, or falsified. They also highlight the need for quantitative formalization, experimental validation, and integration with existing physical theories.

20 Conclusion

The Triadic Cosmos framework provides a unified temporal ontology in which entropy, recursion index, mass, and dual time jointly determine the evolution, stability, and lifetime of physical closures across all scales. The five structural theorems developed in this work establish that aging, material lifetime, and temporal flow are not domain-specific phenomena but universal properties of entropic closures. Biological organisms, chemical systems, planetary bodies, stars, neutron stars, and horizon-bounded objects all obey the same structural laws.

The first theorem shows that aging is the entropic acceleration of material time: as entropy accumulates, the recursion index rises and the internal temporal flow of the closure accelerates. The second theorem demonstrates that this process is structurally inevitable for any active closure. Entropy production is inseparable from participation in the triadic cycle, and aging can be slowed but never eliminated without halting recursion entirely.

The third theorem reveals that material lifetime is a monotonic function of entropy. High-entropy closures evolve rapidly and possess short lifetimes, while low-entropy closures evolve slowly and can persist for cosmological durations. This principle explains the rapid evolution of massive stars, the moderate aging of biological organisms, and the near-frozen temporal behaviour of degenerate matter.

The fourth theorem introduces mass-scaled entropic recursion: the rate at which entropy increases recursion index is inversely proportional to mass. High-mass closures experience large entropic increments

and rapid recursion acceleration, while low-mass closures experience small increments and slow acceleration. This mass-scaled law explains why massive stars evolve quickly, why humans age gradually, why microscopic organisms age extremely slowly, and why large dinosaurs exhibited long lifespans.

The fifth theorem establishes the independence of aging from electromagnetic time. EM processes evolve under t_{EM} and can influence material closures only indirectly through energy transfer or structural change. Aging itself remains a purely material-time phenomenon governed by entropy and recursion index.

Together, these theorems reveal that temporal behaviour across the cosmos is governed by a single structural principle: entropy drives recursion, recursion shapes material time, and mass modulates the rate of entropic impact. Aging, evolution, stability, and collapse are therefore unified expressions of the same triadic architecture. The resulting ontology provides a coherent, predictive, and empirically grounded interpretation of cross-domain temporal phenomena, linking biological aging, planetary cooling, chemical maturation, stellar evolution, neutron-star longevity, and horizon-bounded dynamics under one universal temporal framework.

A Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

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